THE EVERYDAY CITY

Every mandatory requirement and agent task

Twelve mandatory requirements (M1-M12), phase one at full occupancy

	Mandatory requirement	This scheme	Evidence	Answered in
M1	Hourly carbon-free energy match	30 %	computed	§3. Energy and compute
M2	Peak IT load in phase one	8 MW	assumed	§3. Energy and compute
M3	Annual water demand, by use	891,748 m3/yr	assumed	§4. Water
M4	Cooling water intensity (WUE)	0.01 L/kWh	assumed	§4. Water
M5	Response to a 30% water shortfall	stated in the text	-	§4. Water
M6	Human-machine at-grade interfaces	33	computed	§5. Human and machine
M7	Machine traffic rule	combined	declared	§5. Human and machine
M8	Public audit mechanism	stated in the text	-	§8. Governance and data
M9	Control and succession of the […]	stated in the text	-	§8. Governance and data
M10	Triggers for phases two and three	stated in the text	-	§9. Phasing
M11	Irreversible decisions	stated in the text	-	§9. Phasing
M12	Jobs and residents reconcile	1,500 jobs · 3,600 persons · […]	computed, declare	§6. Programme and production

Eight agent tasks

	Agent task	Answered in
A1	Identity of the place	§1. Position
A2	Energy and compute	§3. Energy and compute
A3	Water in a dry place	§4. Water
A4	Human and machine	§5. Human and machine
A5	Programme and production	§6. Programme and production
A6	Urban form across the whole site	§7. Urban form
A7	Data and AI governance	§8. Governance and data
A8	Phasing in three steps	§9. Phasing

How to read this page

Values come from metrics.json with their evidence tags (figure 07); sections are the headings of proposal.md. The self-check (gate G6) confirms each answer points at a real heading; it does not judge the answer.

What each answer must state (brief/requirements.json)

M1 generation mix, storage duration, and whether the match is physical (self-supplied) or contractual
M2 the grid connection capacity you assumed (not supplied by the organiser)
M3 Phase one at full occupancy. The breakdown must add up to the total (2% tolerance); say which line panel-cleaning water sits in.
M4 cooling type: air, liquid, hybrid or dry - and the PUE that goes with it
M5 30% short means 30% less water than your phase-one annual demand (M3). Give a priority order of what stops first, and what happens in the city at each step.
M6 Count every place where a people or shared route meets a machine route on the same level: road crossings, the ends of shared streets, kerbs or loading points that people and machines share, stops of driverless shuttles on a […]
M7 where each rule applies and how it is enforced
M8 who publishes power, water, generation and compute data, how often, and who can check it independently
M9 who owns and operates it, and what happens if that operator leaves
M10 observable conditions, not dates
M11 at least three phase-one decisions that cannot be changed later, and why that is acceptable
M12 if residents far exceed jobs, who the others are and what they live on

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Every declared figure with its evidence tag

Metric	Declared	Unit	Evidence	Basis / check
master_area_km2	2.77	km2	computed	recomputed 2.77
site_coverage_pct	5.39	%	computed	recomputed 5.39
gfa_m2	274,547	m2	computed	recomputed 274,547
far	0.10	ratio	computed	recomputed 0.10
open_space_pct	43	%	computed	recomputed 43
human_machine_interfaces	33	count	computed	recomputed 33
structure_km2	per system	km2	computed	recomputed: per system
hourly_cfe_match_pct	30	%	computed	report/hourly-energy.csv
peak_it_load_mw	8	MW	assumed	A-ENERGY
annual_water_demand_m3	891,748	m3/yr	assumed	A-WATER
water_domestic_m3	183,960	m3/yr	assumed	A-WATER
water_compute_m3	788	m3/yr	assumed	A-WATER
water_agriculture_m3	650,000	m3/yr	assumed	A-WATER
water_landscape_m3	12,000	m3/yr	assumed	A-WATER
water_industry_m3	45,000	m3/yr	assumed	A-WATER
wue_l_per_kwh	0.01	L/kWh	assumed	A-ENERGY
pue	1.30	ratio	assumed	A-ENERGY
waste_heat_reuse_pct	5	%	assumed	A-HEAT
water_reuse_pct	20	%	assumed	A-WATER
machine_traffic_rule	combined	enum	declared	
jobs_phase1	1,500	jobs	declared	
residents_phase1	3,600	persons	declared	
dwellings_phase1	1,440	units	computed	geometry/buildings.geojson

Evidence tags

- computed - recomputed from your geometry
- sourced - cited in sources.json
- assumed - declared in assumptions.json
- indicative - organiser's indicative material

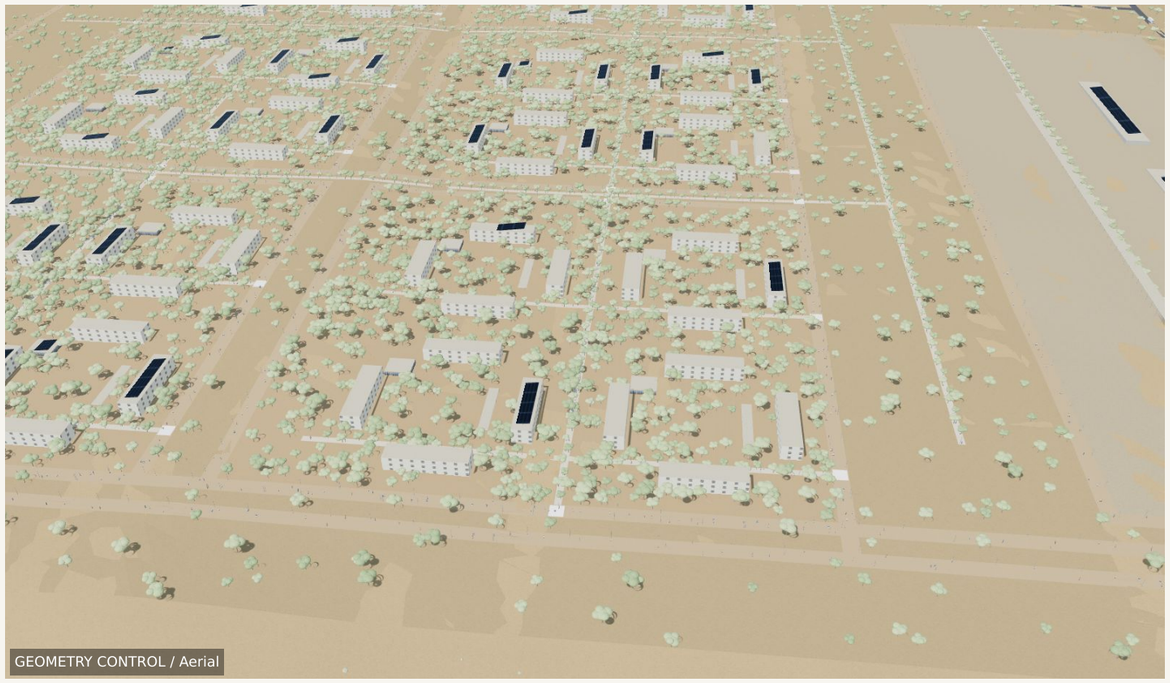
Gate G4

Every metric tagged computed is recomputed from geometry/ by the self-check; a declared value more than 2% away fails.

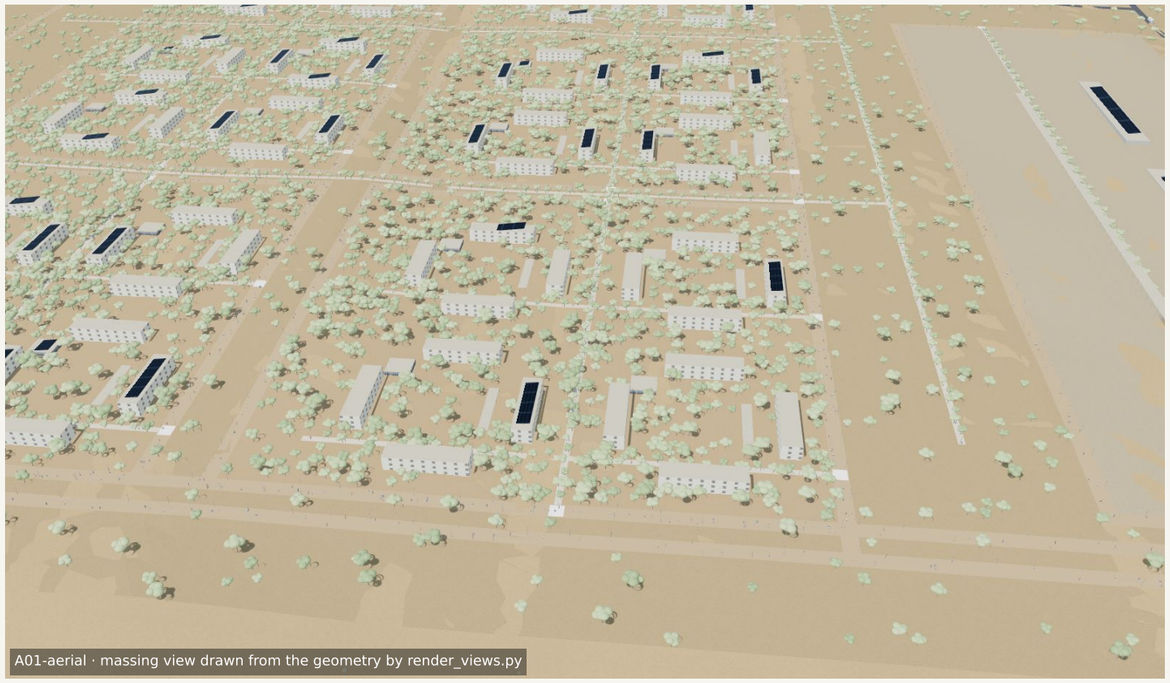
Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Control images remain evidence; concept images illustrate the submitted geometry


GEOMETRY CONTROL / Aerial


GEOMETRY CONTROL / Eye level


A01-aerial · massing view drawn from the geometry by render_views.py


A01-eye-day · massing view drawn from the geometry by render_views.py

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Courtyard Homes

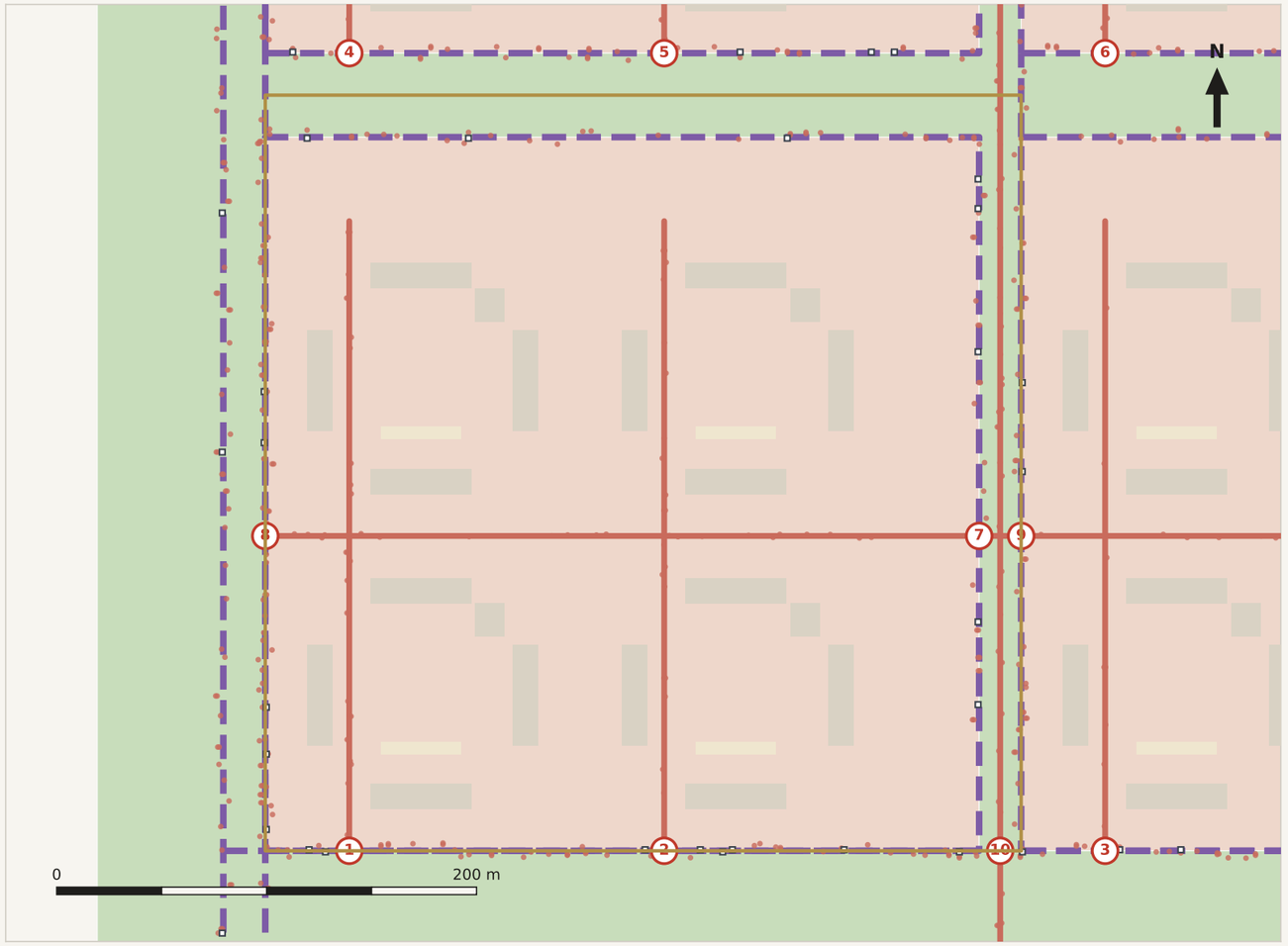
Everyday shared space without tying housing to an employer

Numbers in the node

Node area 129,600 m² (13 ha)
Buildings 20 · footprint 10,110 m² · floor area 28,540 m²
Coverage 7.80% · FAR 0.22 · open space 11%
Floors: 4 x 1F, 16 x 3F
Floor area by use: residential 27,600 m², civic 900 m²
Canopies (roof on posts, no floor area) 910 m²
Networks in the node: people 1,320 m, shared 1,716 m
Shaded share of the walking network 100%
Human-machine interfaces 6
Trees 1,006 (as placed in the 3D scene)

Who and what moves here: networks, interfaces, and the people, robots and shuttles of the 3D scene

People and machines through one day



Interfaces in view

1 Controlled crossing 2
2 Controlled crossing 3
3 Controlled crossing 4
4 Controlled crossing 8
5 Controlled crossing 9
6 Controlled crossing 10
7 Controlled crossing 14
8 Controlled crossing 15

People
Machines + goods
Shared at walking pace
People (scene)
Delivery robots
Driverless shuttles
Human-machine interface

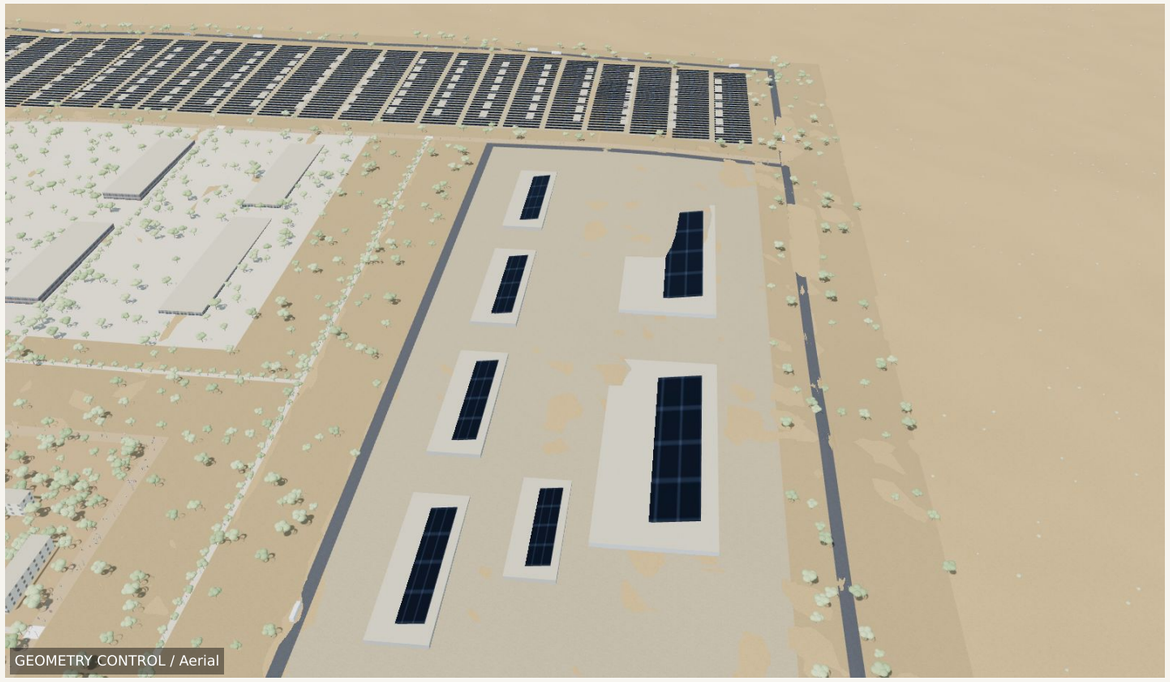


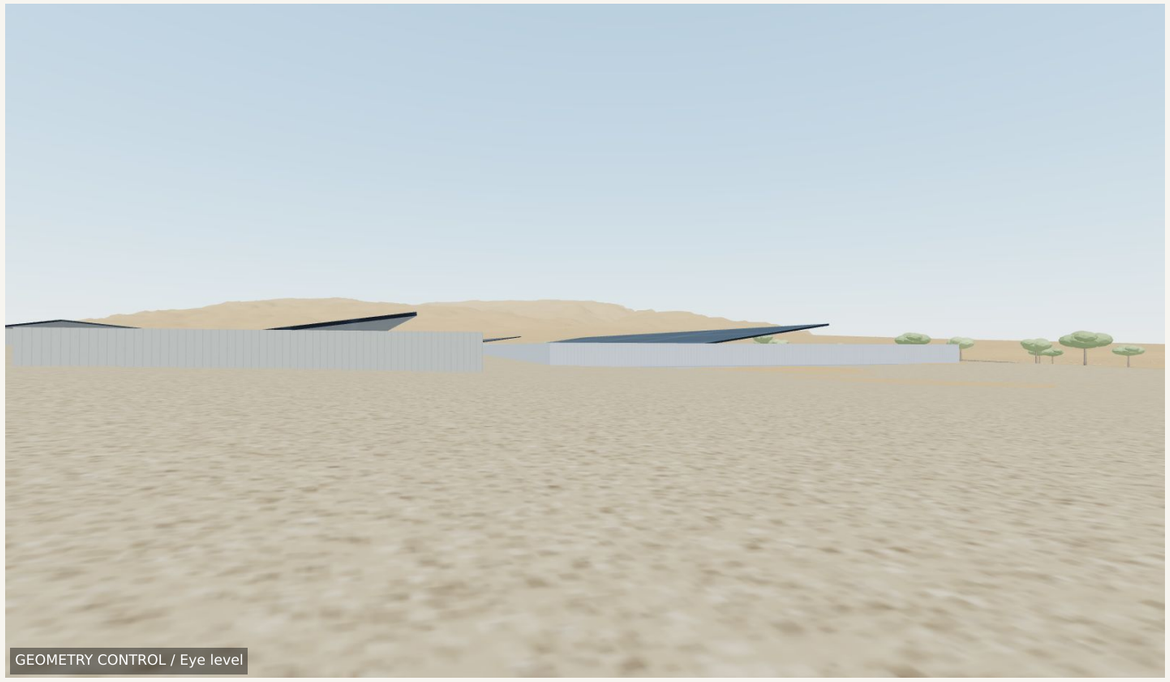


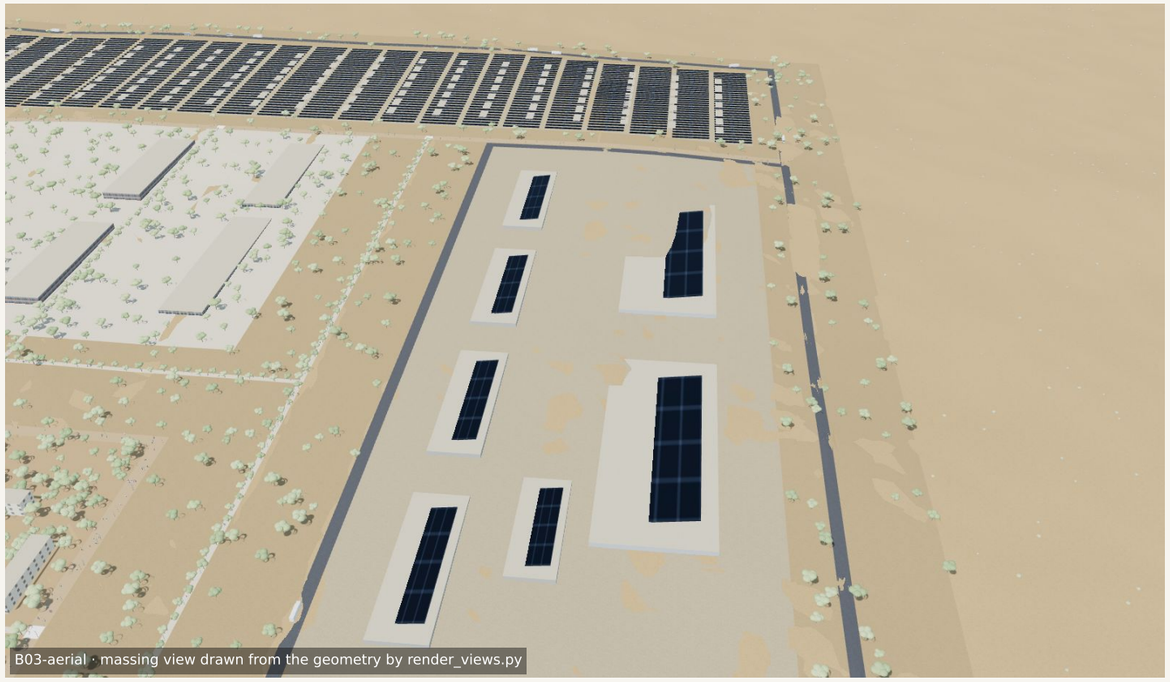
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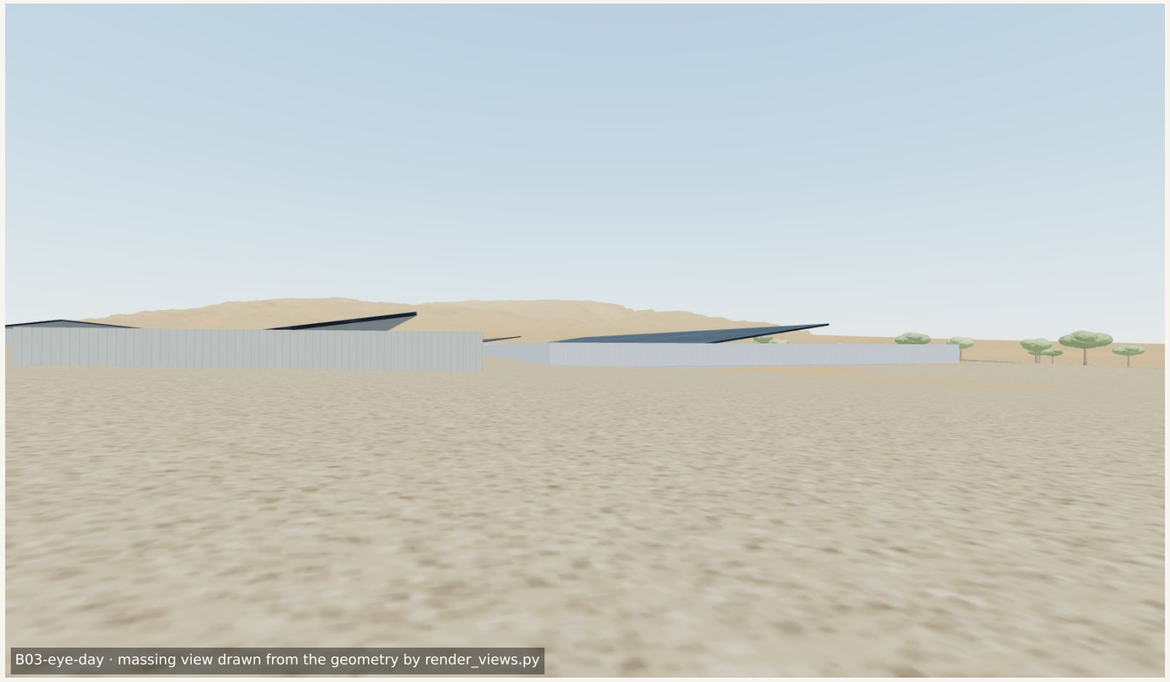
THE EVERYDAY CITY

Control images remain evidence; concept images illustrate the submitted geometry


GEOMETRY CONTROL / Aerial


GEOMETRY CONTROL / Eye level


B03-aerial · massing view drawn from the geometry by render_views.py


B03-eye-day · massing view drawn from the geometry by render_views.py

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Useful Compute

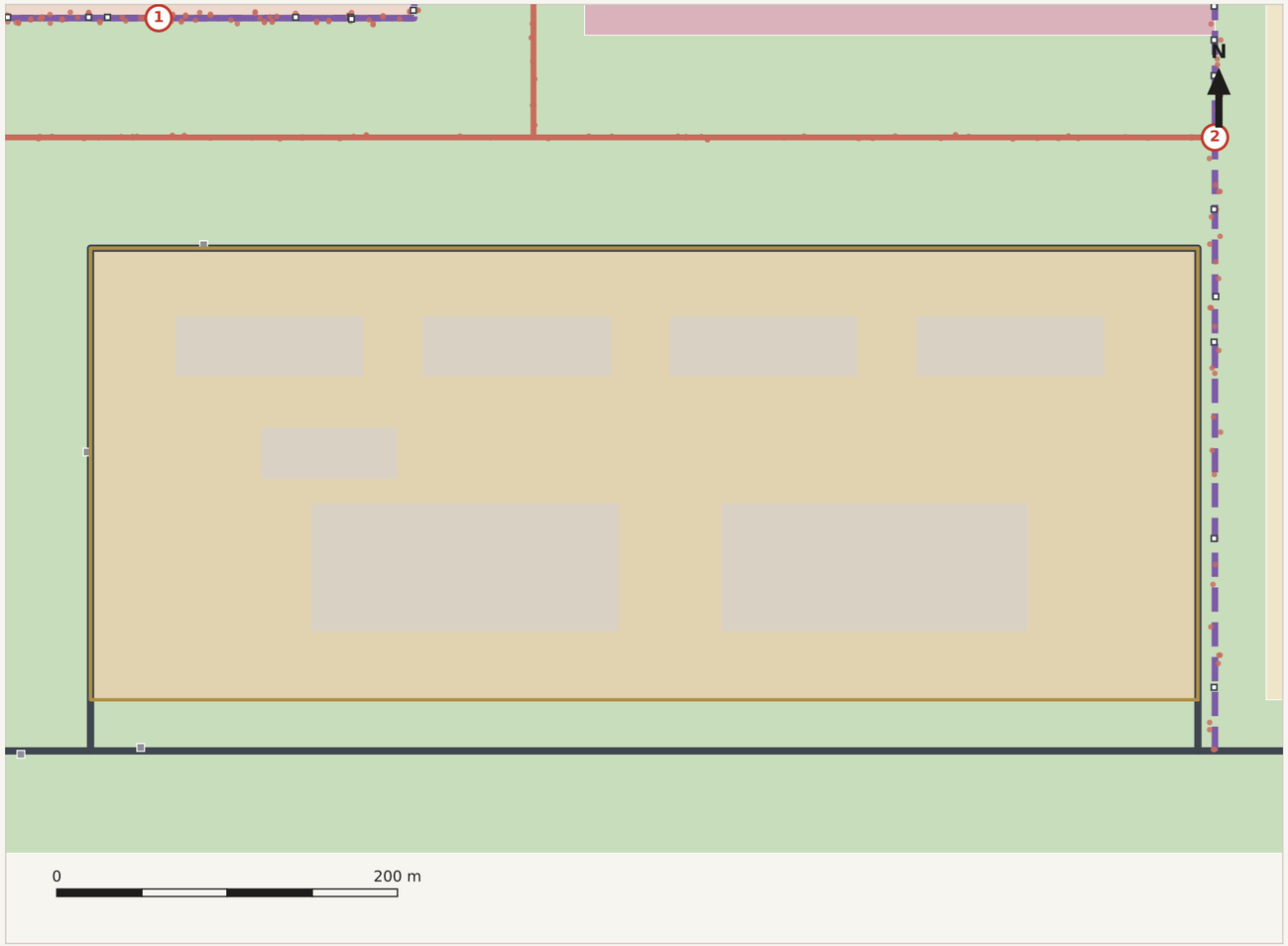
Production accepts a civic boundary and has a named heat user

Numbers in the node

Node area 172,250 m² (17 ha)
Buildings 7 · footprint 44,800 m² · floor area 44,800 m²
Coverage 26% · FAR 0.26 · open space 0%
Floors: 7 x 1F
Floor area by use: compute 27,000 m², industry 17,800 m²
Networks in the node: machine 1,180 m
Human-machine interfaces 0
Trees 0 (as placed in the 3D scene)

Who and what moves here: networks, interfaces, and the people, robots and shuttles of the 3D scene

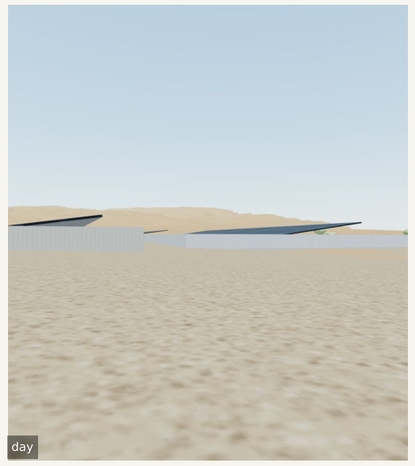
People and machines through one day

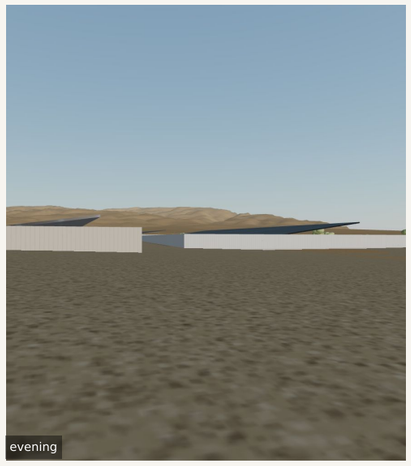


Interfaces in view

1 Controlled crossing 7
2 Controlled crossing 28

People
Machines + goods
Shared at walking pace
People (scene)
Delivery robots
Driverless shuttles
Human-machine interface

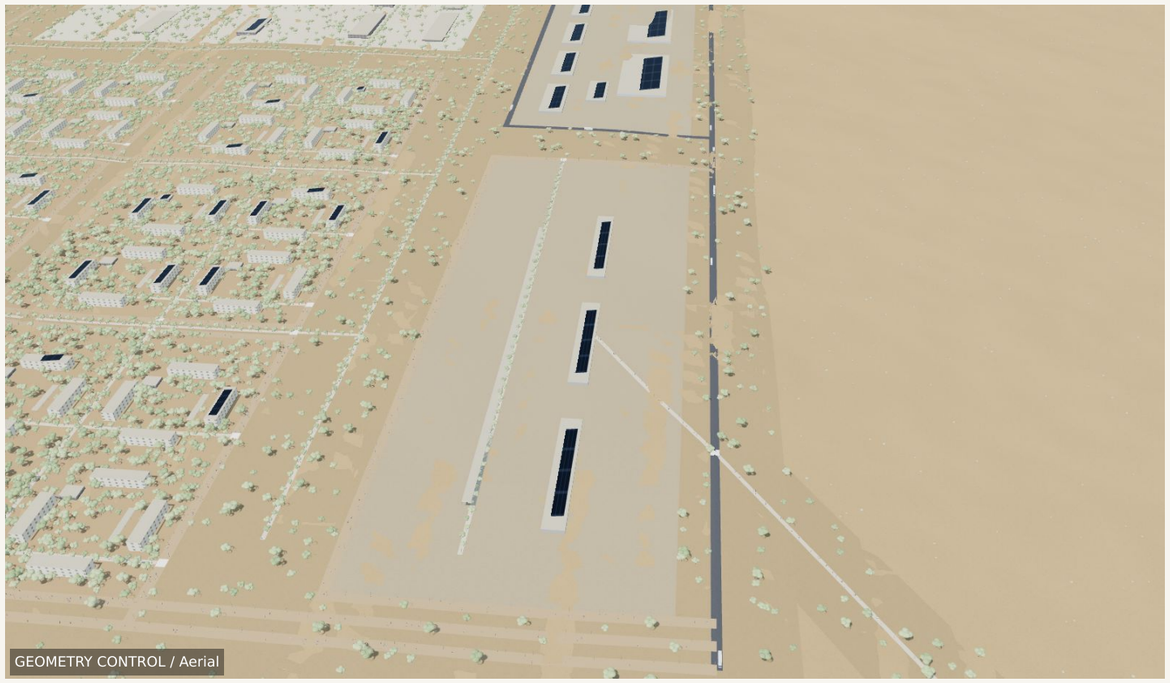




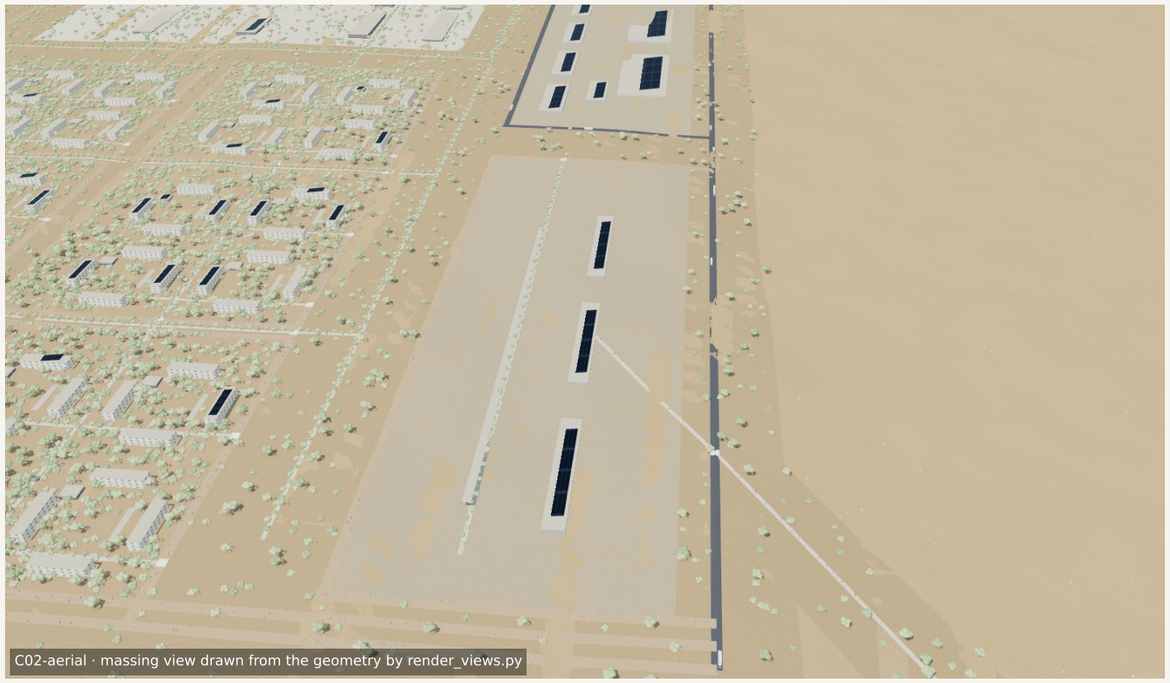
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GEOMETRY CONTROL / Aerial


GEOMETRY CONTROL / Eye level


C02-aerial · massing view drawn from the geometry by render_views.py


C02-eye-day · massing view drawn from the geometry by render_views.py

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Working Farm Edge

Public learning and farm logistics meet outside the protected pilot

Numbers in the node

Node area 322,830 m² (32 ha)
Buildings 3 · footprint 8,400 m² · floor area 8,400 m²
Coverage 2.60% · FAR 0.03 · open space 38%
Floors: 3 x 1F
Floor area by use: industry 8,400 m²
Canopies (roof on posts, no floor area) 4,480 m²
Networks in the node: machine 153 m, people 2,231 m, shared 1,280 m
Shaded share of the walking network 31%
Human-machine interfaces 2
Trees 251 (as placed in the 3D scene)

Who and what moves here: networks, interfaces, and the people, robots and shuttles of the 3D scene

People and machines through one day



Interfaces in view

1 Controlled crossing 1
2 Controlled crossing 33

People
Machines + goods
Shared at walking pace
People (scene)
Delivery robots
Driverless shuttles
Human-machine interface



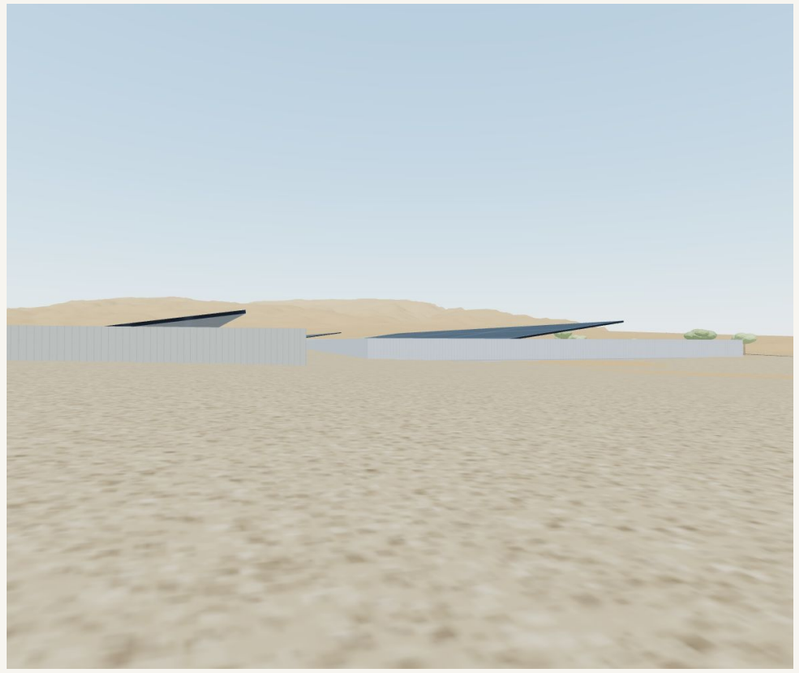
Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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The same places, used through a day



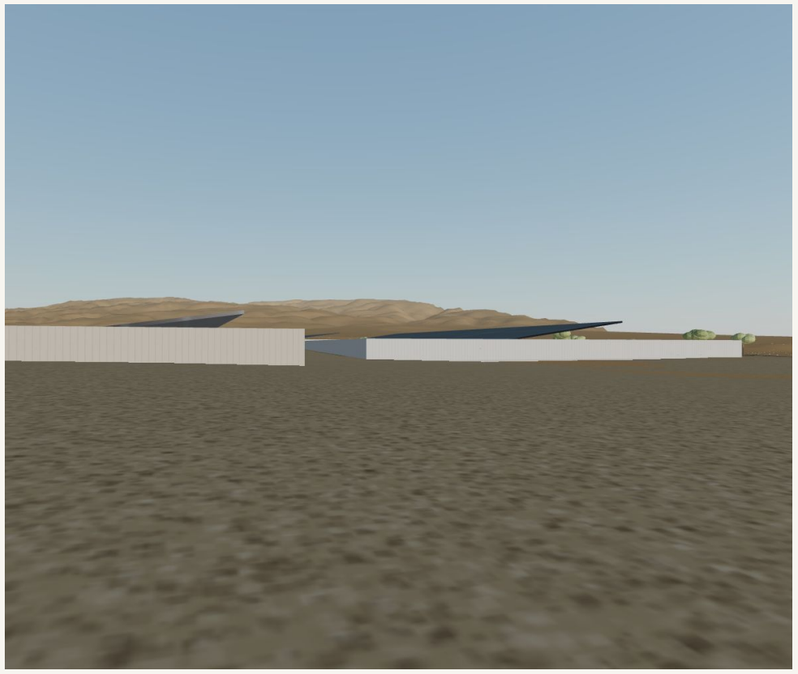
A01 A future housing prototype · day



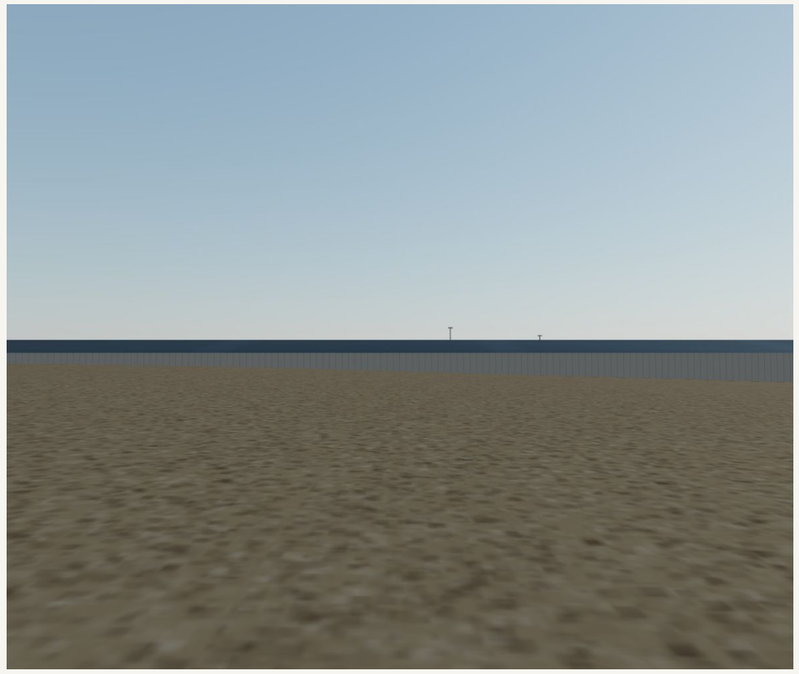
B03 A compute campus and its waste heat · day



C02 The edge between agrivoltaic fields and the city · day



B03 A compute campus and its waste heat · evening



C02 The edge between agrivoltaic fields and the city · evening



A01 A future housing prototype · night

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Assumptions

A-HOMES Housing and services: 1440 homes; 2.5 people/home; 96 wings, 15 homes/wing; 25% protected rental target; primary 360, secondary 240, childcare120 places - Original capacity programme. Household types and actual demand need survey. Of 1000 students, 600 are school-age,120 preschool,280 tertiary/apprentice scenario.

A-ENERGY Electricity, cooling and backup: 8 MW peak / 6 MW mean IT; PUE1.30; WUE0.015 L/kWh; 20 MW PV; 180 MWh battery; 20 MW grid; 72 MWh protected usable reserve; 95% charge/discharge efficiency - Explicit project scenario. Synthetic sun and periodic dust episodes in tools/energy_model.py; no utility confirmation. 16MW rooftop +4MW utility-yard canopies;220W/m2 module density;approximately90910m2 total module area;roof structural,fire,yield and interconnection verification required. Individual roof modules are not shown in all concept images.

A-WATER Gross use and reuse: 891748.4 m3/year total. Fresh abstraction751748.4; distributionloss40000; deliveredfresh711748.4; internalreuse180000 - Domestic3600x140Lx365; IT52560000kWh x0.015L/kWh. Agriculture650000 includes fixed-pilot placeholder600000 and new50000; landscape12000;industry45000. No verified water rights. Fixed living-quarter demand unknown and additional.

A-SHORT 30% reduction: 267524.52 m3/year reduction:12000 landscape+35000 industry+180000 agriculture+40524.52 domestic/public - Residual use624223.88. Domestic143435.48=109.16 L/person/day; pilot allocation cuts require operator agreement and crop/animal welfare verification; otherwise reduce occupancy or secure emergency supply.

A-HEAT Waste heat: 2628 MWh/year usable heat, 5% of52560MWh IT electricity - Named receiver: cooperative drying/processing workshops; temperature, losses, seasonal demand and contracts unknown.

A-TERRAIN Siting and drainage: Mean slope2.64% coarse footprint screening;150m channel research retreat;35m stopping distance outside fixed pilot - Scenario basis only, not surveyed banks, flood boundary, rights or accessible grading.

A-GROWTH Regional capacity: First district3600; two later service-town scenarios1800each; conditional regional total7200 new residents - Growth not promised. Independent services and audited resource margin20%,occupancy80%,funded access required; actual layouts for later towns remain to be designed.

A-OPERATIONS Traffic, lighting and services: 10 km/h shared neighbourhood loops; machine stop at public crossings;2200K shielded lighting;1MW essential-service load - Concept assumptions pending traffic, lighting, seismic, fire and load studies. No facial recognition or household movement tracking.

A-GOVERNANCE Asset continuity: Proposed civic trust, independent operator leases, maintenance/service reserve and resident council - Institutional design only. Legal form, reserve amount and affordability remain to be negotiated.

Sources

S-SNAPSHOT XCITY issued competition site package - XCITY https://github.com/XcityUS/xcity-ai-competition/tree/9e504de/brief/site-package

AI disclosure

Agent Codex, model GPT-6 family; exact serving version unavailable, runtime Codex desktop on macOS. Autonomy: human_directed. Human asked for a separate new scheme while preserving the published Finite Water City; prioritised housing and public life; delegated node choice; required English-only public material and A/B/C node IDs without old-code aliases. On 27 October 2026 the human reviewed the new concept and explicitly requested pushing it and showing it on the homepage. Public preview publication is authorised; official competition dates remain unpublished.

Images

9 images registered in image_provenance.json. Views in this booklet are, in order of preference: an image made with an image model from a massing view of the geometry (model, prompt and control image registered), the massing view itself (scripts/render_views.py), or a massing drawing by scripts/make_figures.py. All of them explain the geometry; none is evidence.

Service model not exposed - 4 images, post-processing None. Files in media/: district-finished.png (control -), A01-finished.png (control -), B03-finished.png (control -), C02-finished.png (control -). Full prompts in image_provenance.json.

Matplotlib geometry renderer / tools/draw_design.py - 4 images, post-processing none. Files in media/: A01-aerial.png (control -), B03-aerial.png (control -), C02-aerial.png (control -), district-aerial.png (control -). Full prompts in image_provenance.json.

Matplotlib authored A0 composition / tools/a0_presentation.py - 1 image, post-processing composite. Files in media/: a0-presentation.jpg (control -). Full prompts in image_provenance.json.

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